

1 Operation of AIP

1.1 Special control and display elements within AIP

Tables

Tables are displayed in a uniform way within AIP. This affects the basic display (workplaces, operations, ...) as well as the selection lists of posting dialogs.

Filter	
Status	Status
1	PRODUCTION
2	SETUP
3	ORDER SHORTAGE
4	OUT OF MATERIAL
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE

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Provided that information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. By clicking/touching the user can directly switch to another page.

An operation may be selected using the mouse, touch screen, keyboard (arrow keys: '↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Please find the following example: When an operation is logged on, those operations may be selected that are included in the sequencing list or that are planned for the corresponding workplace or group. However, when operations are interrupted, only running operations may be selected.



Scrolling page by page (up or down) in the table.



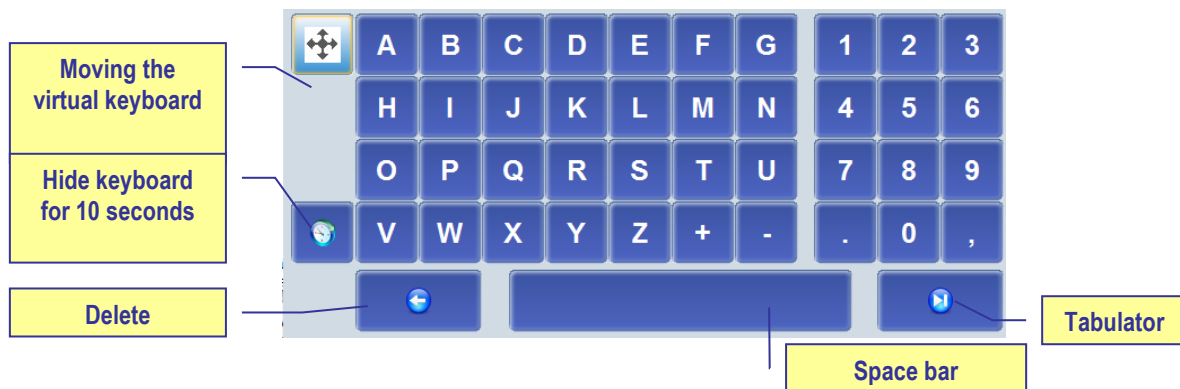
Scrolling to the left or right. Only those buttons are activated that are reasonable for the current situation. This figure shows that scrolling to the left has been deactivated.

Filter A

“table filter” may optionally be displayed (customizing). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is case-insensitive.

Virtual keyboard

The virtual keyboard allows for data to be entered manually via touch screen or a connected mouse. To make it easier for inexperienced users to find the required keys, the numeric key pad is organized like the telephone and letters are aligned in alphabetical order. Consequently, both differ from the computer keyboard which usually is aligned in the “QWERTZ keyboard layout”. The virtual keyboard is displayed automatically as soon as an input field is focused.



The driver needs to be configured respectively for the touch screen to be able to move the keyboard (settings in the control panel of the terminal/PC)! The virtual keyboard only supports the characters "0" - "9", "A" - "Z" and "+", "-", ".", and ",". Other characters or languages are not supported. It is recommendable to use an additional keyboard if texts in other languages have to be entered.

The start position of the virtual keyboard can be defined by a setting in the configuration file keyboard.ini. Subject to the screen resolution, the parameters xpos= and ypos= need to be enabled in the configuration file.

If the virtual keyboard is not to be shown in general, the parameter -t needs to be included in the parameter bar parameters= of the configuration file ctaip.ini.

Date display

AIP supports a country-specific date format in dynamic dialogs. This can be configured in the "control panel", "regional settings", "short date" dialog of the terminal/PC. The following has to be taken into account in this context:

- Years are always four characters long.
- Months and days are always 2 characters long.
- “-”, “/” and “.” are allowed separators
- Blanks must not be included in the “short date” format, i.e. the <BLANK> separator is not allowed.
- The date separator “.” (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- | | |
|-----------------------|------------|
| • English(USA) | MM/DD/YYYY |
| • Danish | MM-DD-YYYY |
| • Customer-specific 1 | YYYY-MM-DD |
| • Customer-specific 2 | YYYY/MM/DD |
| • Customer-specific 3 | MM/YYYY/DD |

Please note

If the date format does not correspond to conventions a note appears when the program is started and the date format is set to MM/DD/YYYY.

The year is displayed only by two characters in the status bar.

1.2 General description of the posting process with AIP

In general, posting dialogs are divided into several visual views at AIP. These views (partial dialogs) cover the entire screen and only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. This process is described by way of the following example (interrupt operation). The other dialogs can be operated in the same way.

The “interrupt operation” function is executed. This task is started by clicking the “interrupt operation” function from the second toolbar:



The “interrupt operation” dialog opens and the first view is displayed. The function that is currently being executed (in this case: interrupt operation) is shown in the header.

1st view (partial dialog)
The views are consecutive

Posting function that is currently being executed

Interrupt operation S30002090200 Deburring at workplace 33021

1 Enter quantities 2 Choose status 3 Confirmation

Operation 0200 Deburring

Prod.Order S30002090200

Article SAR-VPO011-M-9005

Mirror housing right, bla

Target qty. 1200 PCS

Yield 660 PCS

Scrap 0 PCS

Completion 55%

Quantities that have already been collected (yield, scrap)

General OP data

Yield 22

Scrap 7

Active input field

Virtual keyboard

Cancel Go on

mpcv 07/12/12 23:08:28

The first view “enter quantities” provides the user with the possibility to enter the produced yield or scrap quantities. The virtual keyboard is shown or hidden automatically, subject to the active input field.

Quantities can be entered using the virtual keyboard or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). Once all values have been entered in the first view, the next view can be opened by clicking the “next” button.

The “cancel” button is displayed in all partial dialogs and allows for the entire posting dialog to be cancelled/closed at any time.

The next view can be opened either by clicking the “next” button or by clicking another tab (in our example: “select status” or “confirm”). Please note in this context, that no view can be skipped when the views are navigated bottom up (view 1 → view 2 → view 3). This means: if you are in the first view (enter quantities) and you click the third view (confirm), the second view (select status) will be displayed first.

Vice versa, when navigating top down (e.g. from the “confirm” view to the “enter quantities” view), every view may directly be opened by clicking at it. In this case, views are actually skipped. But the “back” button also allows for the views to be opened one after the other (top down).

As long as the dialog has not been confirmed, entered data may be changed at any time by scrolling back and forth.

AIP

Interrupt operation 4199LL120010 Assembly at workplace 406457A

1 Enter quantities 2 Choose status 3 Confirmation

Workplace 406457A MOUNT1

Actual status PRODUCTION since .. 18:40:41

Filter

Status	Status
1	PRODUCTION
5	OUT OF STAFF
6	START UP
22	COLOUR CHANGE
97	ELECTRICAL FAILURE
98	MECHANICAL FAILURE
99	GENERAL DISTURBANCE
501	COLOUR SAMPLE

New status 6 START UP Estimated duration 0 Minutes

Comment

Cancel Back Go on

mpcv 07/12/12 23:14:58

The workplace status that is to be set, once the operation has been interrupted, is determined in the second view “select status”. This status may be chosen from the displayed status list. This list can be restricted using the “filter” field. Once the required values have been entered, the next view can be opened by clicking “next” (in our example it is the last view).

Interrupt operation 4199LL120010 Assembly at workplace 406457A

1 Enter quantities 2 Choose status 3 Confirmation

Operation 0010 Assembly

Prod.Order	4199LL12	Target qty.	30000 PCS
Article	34364-647-345	Yield	6833 PCS
	Seating Plate	Scrap	333
		Completion	22%

Workplace 406457A MOUNT1

New status	START UP
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Workplace data

Entered quantities

Yield	22
Scrap	7

Quantities posted for the OP.

Staff Number

Input field for the badge number

Cancel

Back

Interrupt OP

07/12/12 23:16:40

The partial dialog “confirm” shows a summary of all values entered so far in the dialog. Provided that the user agrees with the entered data, the “interrupt operation” dialog can be confirmed, once the badge number has been entered. Then the dialog including the data is sent to the server and posted.

If an input field of the dialog is not filled out properly (e.g. a mandatory field is empty) the field is highlighted in red in the corresponding view and focused to enable the user to directly correct the field content.



If a workflow dialog is opened it may directly be exited by clicking the ESC button. This is also the case, if the virtual keyboard is opened. Thus, the ESC button cannot be used to close the virtual keyboard.